CHAPTER-7

SOFTWARE REQUIREMENTS

7.1 Design Tools:

There are 3 tools that are used for this process

- S-edit
- T-SPICE L-edit
- A schematic capture tool
- The SPICE simulation engine integrated with S-edit-the physical design tool

7.2 Usings-Edit (Schematic Entry Tool) & T-Spice (Analog SimulationTool):

PART1: Set up and download your libraries directory structure

- Log on the sixth level of a Coble high computer.
- For all your Tanner EDA projects, you want to build a directory. You must also download and disassemble a set of library & model files for your simulations from the course website.
- Create the "EELE414 VLSI Fall2011\Tanner_ Projects" directory structure
- Go to the webpage for the course and download "Tanner Library. Zip" zip file. Distributed to your directory Tanner Projects. This collection of files contains information required for S-edit components (circuit symbols) to be simulated by SPICE.

PART2: Launching a new design and installation library

- S-Edit Start: The EDA Tanner -Tanner Tools v12.6 - S-Edit v12.6
- New design launch: Create a new design using pull-down menus:
- New file design
- New file

A dialogue requesting the design name and placement will display. When named, S-edit creates a folder in the location you supply containing all the design files for that name. You should name each simulation you are conducting in a description. A cell is an element of design.

There may be many perspectives of a cell such as schemes and symbols. Other cells may be instantiated. If you simulate, usually, if you test a circuit, we name your cell "TOP" – for example, an inverter, which has its own cell, a device scheme and a symbol. The TOP cell containing ideal components, such as voltage sources and probes, needed exclusively for simulation, is instantiated into the inverter cell. It helps us to differentiate the mounted cells from the cells that are now utilised simply for simulation on the die versus cells.

Create a new cell view using the pull-down menu:

- Cell– New View:

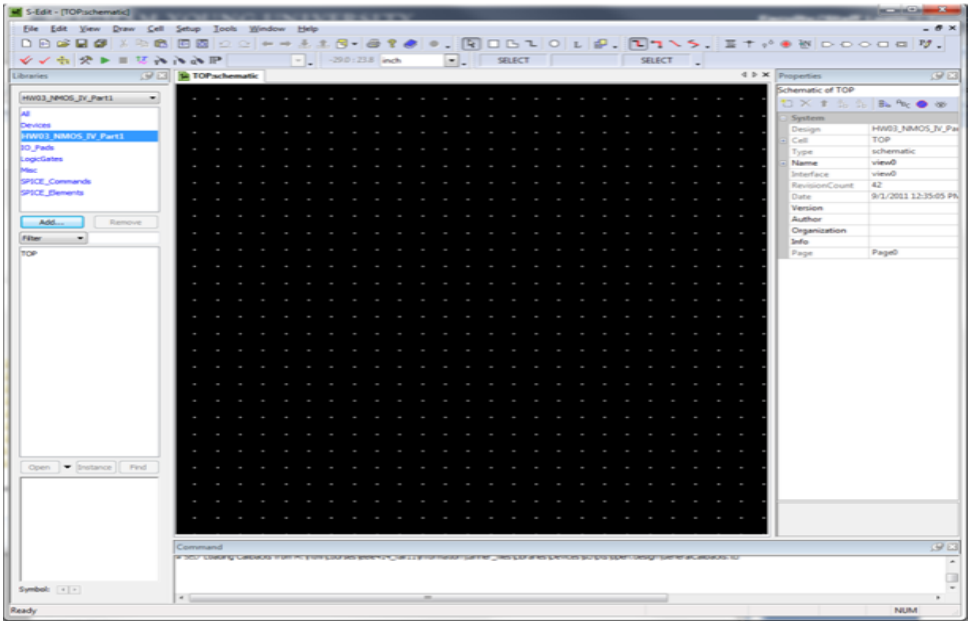

Fig.7.1: Cell New View

- Enter the cell name “TOP”. Ensure the design name is “HW03_NMOS_IV_Part1” andclick OK. You can leave the interface and view names “view0”.

A blank schematic page will appear. It is a good idea to save this right now.

7.3 Enter the Symbol Libraries:

First, you need to include a *library* which contains the symbols for all basic circuit elements such as resistors, NMOS, capacitors, etc… The libraries for all the basic symbols are in the Tanner_Libraries.zip file you downloaded and unzipped.

- On the left side of the S-edit screen you’ll see a *Libraries* window, click on the “Add”button.

- Browse to "Libraries\All\All.tanner" and click "OK" You should see a set of libraries appear:

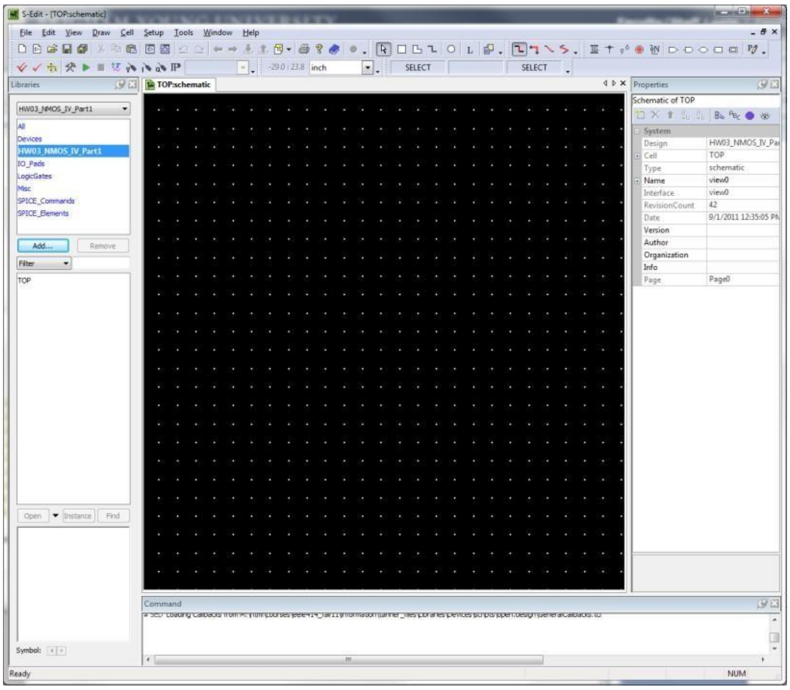

Fig.7.2: Symbol Libraries

7.4 Setup the SPICE Models for the Generic_025 Kit:

The libraries that you just added have symbols for NMOS and PMOS transistors. However, all non-linear components such as MOS transistors require a model to describe their behavior. If you simply enter an NMOS symbol in your schematic, SPICE will not know what to do since each NMOS transistor fabricated in a different technology behave differently.

In this example, we will use a transistor technology called "Generic_025", which represents a standard, 0.25um CMOS process. You will need to setup the SPICE models for this process in S-edit. Once you do that, when you enter an NMOS or PMOStransistor, you can then associate the 0.25um model to that symbol.

Using the pull down menus, setup the SPICE models:

- Setup – SPICE Simulation
- In the dialog that appears, you should highlight "General" on the left.
- On the right, click in the "SPICE File Name" field. This is where you specify the name and location of the SPICE Net-list output. Browse to your design directory "EELE414_VLSI_Fall2011\Tanner- Projects\HW03_NMOS_IV_Part1"and enter the

filename “TOP.SP”.

- On the right, click in the ”simulations results file name ”field this is where the results of the simulation will be written. this file is what the wave form viewer will look for when you go to plot your results. Browse to your design directory and enter the filename TOP.sP.

Before you can exit this window, you will need to select an analysis type. We will setup the details of the analysis later, but for now, just check the “DC Sweep Analysis” and click “OK” to close the setup window.

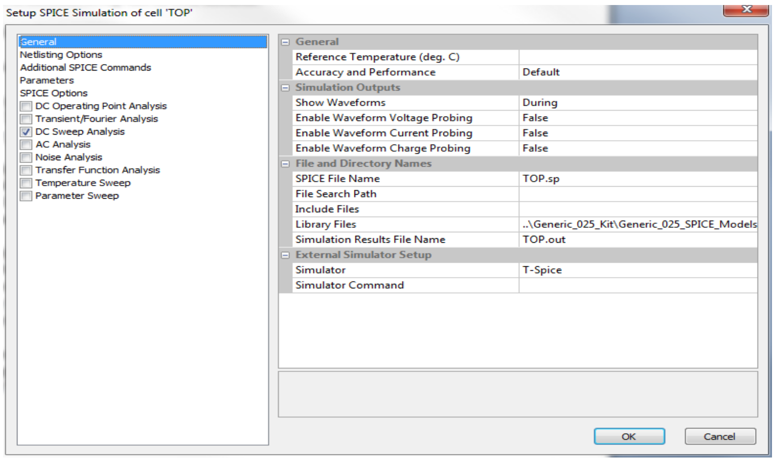

Fig.7.3: Setup the SPICE Models

7.5 Schematic to Simulate the NMOS Transistor:

We will be entering the following circuit.

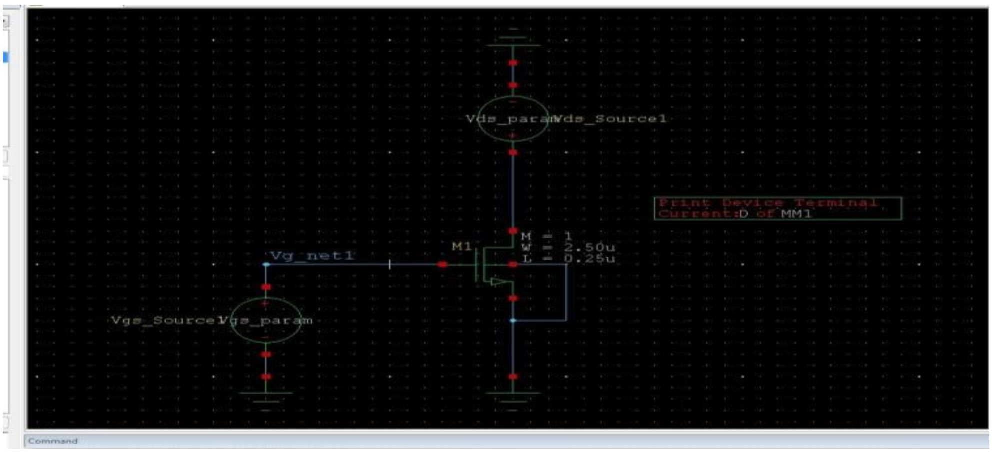

Fig 7.4: Schematic to Simulate the N-MOS Transistor

7.5.1 Enter the NMOS Transistor:

- On the left, click on "Devices" in the upper window. This will display all of the symbolsavailable in this group. You should see all of the components that you can implement on aCMOS integrated circuit.
- On the bottom left window, click once on "NMOS". You should see the symbol of theNMOS transistor show up in the symbol viewer window at the bottom.

To place the NMOS, you will click on the "Instance" button. Two things happen when you click on this button. First, a dialog will appear that will allow you to setup the parameters for the NMOS. Second, the symbol will attach to your mouse. We will placethe NMOS in the schematic first and then set its properties later. This is an easier way to enter the device. Click in the schematic window to drop an instance of the NMOS. Hit the "Esc" button to end the insert-mode.

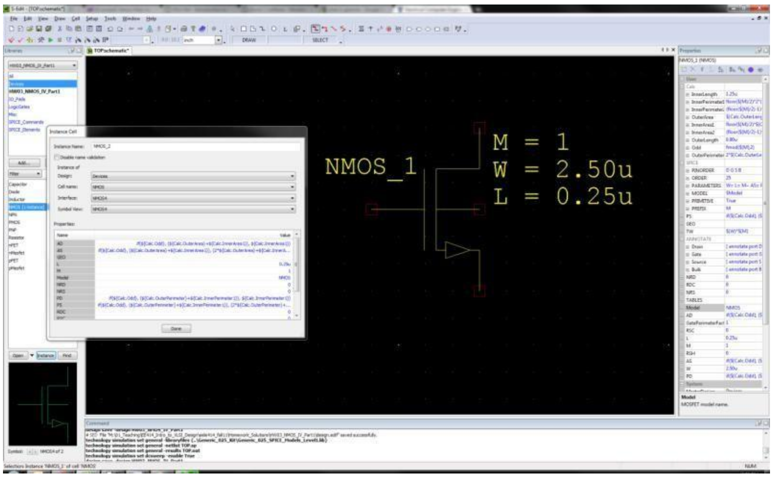

Fig 7.5: N-MOS Transistor

The NMOS is now in the schematic.

A note on zooming:

- [Home] = zoom fit
- [-] = zoom out
- [=] = zoom in
- the scroll wheel also zooms in/out.
- To setup the NMOS, click on the NMOS symbol. You will see the properties of thedevice on the left. We want to setup the following:

- **Model :** enter “NMOS”. This model is found in the Generic_025 library you added.
- **Name:** M1. The SPICE designation for MOS transistors is to have the name start with an “M”. S-edit automatically appends an M to thename is the final name willbe “MM1” in the TOP.sp file. But it isgood practice to name all MOS transistors with M’s.
- W Set to 2.5u. This is the default.
- L Set to 0.25u. This is the default.

7.5.2 Enter a DC source for VGS:

- Using the same process you used for the NMOS symbol, enter “SPICE_Elements:VoltageSource”. This is a generic voltage source symbol that isconfigured as a DC, TRAN, PWL, etc.. in its properties dialog.
- Click on the voltage source and enter the following:
- **MasterInterface:** DC (this is the default but this is how you would change it tosomethingelse.
- **Name:** VGS_Source (it is a good idea to use descriptive names)
- **V**: This is where you will set the DC voltage (i.e., 4v, 5v).

However, for this example we will use a parameter instead of a hardcoded value. We will enter a parameter name here and then set up the parameter later. Enter “VGS_param” for the value of V. When performing a DC sweep, you must useparameters for the sweep.

7.5.3 Enter a DC source for VDS:

- Using the same process as above, enter a DC source for VGS with the following:
- MasterInterface: DC (this is the default but this is how you would change it tosomething else.
- Name: VDS_Source
- V: “VDS_param”

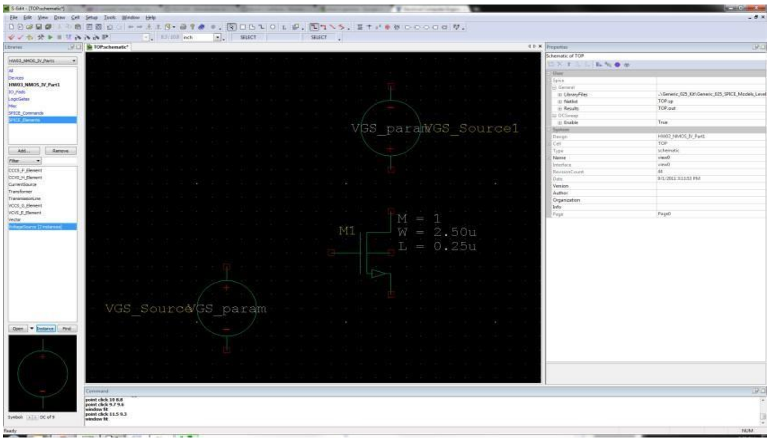

Fig 7.6: Dc Source for VDS

7.5.4 Enter Grounds:

- Using the same process as above, enter 3 grounds from Misc:Gnd

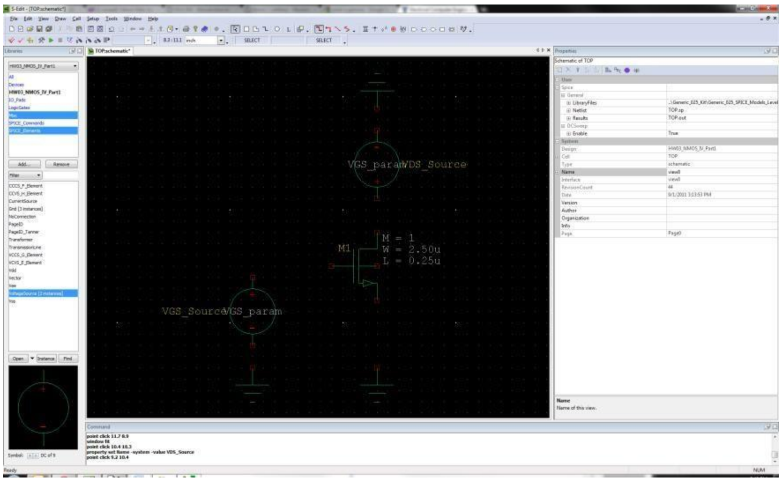

Fig 7.7: Enter Grounds

7.5.5 Enter Wires:

- You can enter wires by clicking on the “wire” icon at the top

Fig 7.8: Enter Wires

- Enter wires by clicking on a symbol node and then dragging. Enter corners by clickingonce where you want to turn.
- You can label nets using the “Net Label” icon at the top.

7.5.6 Enter a Current Probe to Monitor IDS:

- Enter the SPICE_Commands:PrintCurrent component. This doesn’t connect to anything.You just place it anywhere and then tell it what current to monitor in its properties dialog.
- In its properties dialog, setup:
- Terminal: D (this is the Drain of the NMOS)
- Device: MM1 (this is the name of the device. Notice that we called it M1, but S-edit automatically appends another M to the name. You will only see this once you run the Netlist.
- Analysis: DC **(VERY IMPORTANT TO SELECT THIS!!!!)**

7.6 Setup the Parameters that will be Used During the DC SweepAnalysis:

When we entered the VGS and VDS sources, we set their values to “VGS_param” and “VDS_param”.We now need to setup these parameters.

Using the pull down menus:

- Setup – SPICE Simulations
- On the left, click on “Parameters”
- On the right, click on the “Add Parameters” button (it is in the upper right corner next to the red X)
- Enter: Name: VGS_param
- Value: 1v
- On the right, click on the “Add Parameters” button
- Enter: Name: VDS_param
- Value: 2.5v

We will overwrite these values during our sweep, but the parameters need to exist first.

7.7 Setup the SPICE DC Sweep Analysis:

- Using the pull down menus:
- Setup – SPICE Simulations
- On the left, click on "DC Sweep Analysis"
- On the right, enter the following for Source (this is what will be swept)Source or

  Parameter Name: VDS_param

  Start Value: 0

  Stop Value: 2.5

  Step: 0.1

  Sweep Type: lin

- On the right, enter the following for Source (this is what will be swept)Source or

  Parameter Name: VGS_param

  Start Value: 1

  Stop Value: 1.5

  Step: 0.5

  Sweep Type: lin

NOTE: The first parameter you setup in this dialog will be plotted on theindependent axis.

7.8 Simulate the Design:

7.8.1 First, Check you Design Using the Pull Down Menus:

- Tools – Design Checks (any warnings or errors will be shown at the bottom)

7.8.2 Simulate your Design:

- Clock on the Green Arrow to start the simulator:
- The T-Spice window will appear. If everything is OK, the waveform viewer will alsoappear. If everything worked, your waveforms should look like this:

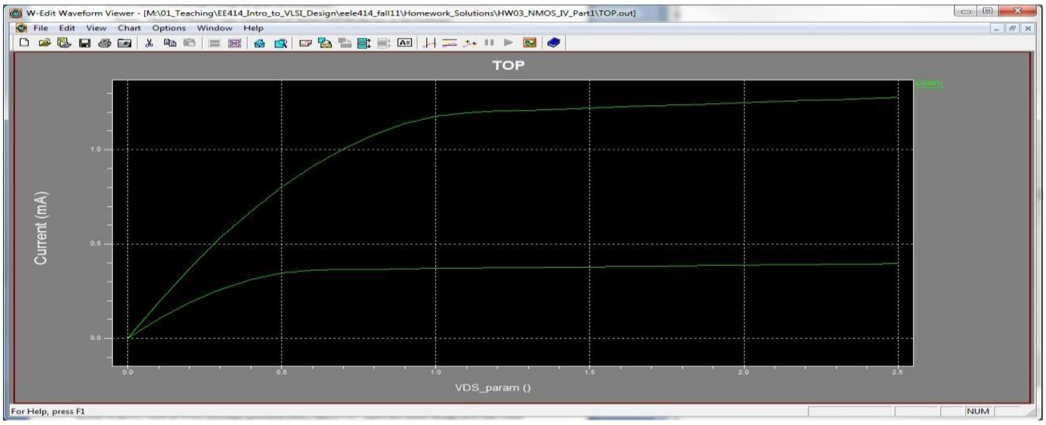

Fig 7.9: Design Simulation

7.8.3 View the Netlist:

- In the T-spice window, right click on the file at the bottom and select "Show Netlist"

This will bring up the TOP.sp Netlist that was created and used by the spice engine. This is a good place to look when you get errors. This is the text based description of what youentered in S-edit.

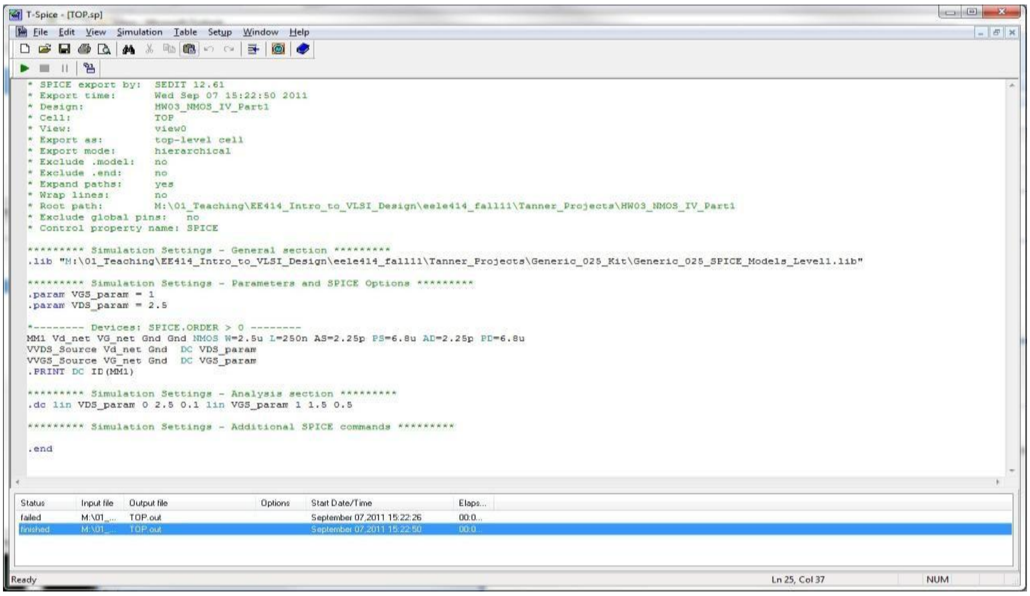

Fig 7.10: View the Netlist

7.8.4 View the Waveform:

- If the windows viewer did NOT automatically appear, you can click on the file inthe T-spice window and select “Show Waveform”

Example: Transient Analysis of a CMOS Inverter & Symbol Creation

7.9 Start a New Design, Setup Libraries & Setup Simulation:

In this example, we will create a CMOS inverter and simulate its transient response. We will create an inverter design that contains a symbol and then instantiate it in another schematic to stimulate the circuit.

Symbols are handled by adding another view to a design. We will start bycreating a design called “Inverter” and then create a schematic view. This schematic will contain an NMOS and PMOS wired as an inverter. We will add “Ports” for the Input, Output, VDD,and VSS. We will then add a symbol view to this design. The symbol will contain the inverter shape and the corresponding pins for Input, Output, VDD, and VSS.

We will then create a separate schematic called TOP that will be used to test the inverter. We will instantiate the inverter symbol in TOP. We only want to put items into the inverter design that can be fabricated. TOP will contain the ideal voltage sources to provide the input waveform, the power supplies, and a mock load. In this way, when we go into physical design (i.e., layout), we only drive forward the isolated circuits.

- Start S-Edit
- Create a new design called:
  “EELE414_VLSI_Fall2011\Tanner_Projects\HW04_INV_Transient_Part1”
- Add the Tanner_Projects\Libraries\All\All.tanner library to the library list on theleft
- Create a new Cell called “TOP” using the Pull Down Menus.
- Cell - New View
- Name = TOP View Type= schematic

- Setup the simulation using the Pull Down Menus:

Highlight the General Tab of the Setup SPICE window and set the following:

SPICE File Name: \HW04_INV_Transient_Part1\TOP.sp

Library Files: ..\Generic_025_Kit\Generic_025_SPICE_Models_Level1.lib Simulation

Results File Name: File Name: \HW04_INV_Transient_Part1\TOP.out

Check the “Transient/Fourier Analysis” box on the left and set the following:

- Stop Time = 2ns
- Maximum Time Step = 10ps

- Click “OK”

7.10 Create the Inverter:

7.10.1 Create a New Schematic View Using the Pull-Down Menus:

- Cell - New View

Name = Inverter

View Type = schematic

7.10.2 Enter the Inverter Schematic:

- Entering the NMOS:

Name = M1

L = 0.25u

W = 2.5u

Model =NMOS

- Entering the PMOS:

Name = M2
L = 0.25u
W = 5.0u
Model =PMOS

Entering the Ports: are entered using the icons on the top of the S-edit window

Enter the following:

- In Port: Name it “IN”
- Out Port: Name it “OUT”In/Out Port: Name it “VDD”In/Out Port: Name it “VSS”

Wire up the Inverter

Enter wire connections as shown in the previous figure.

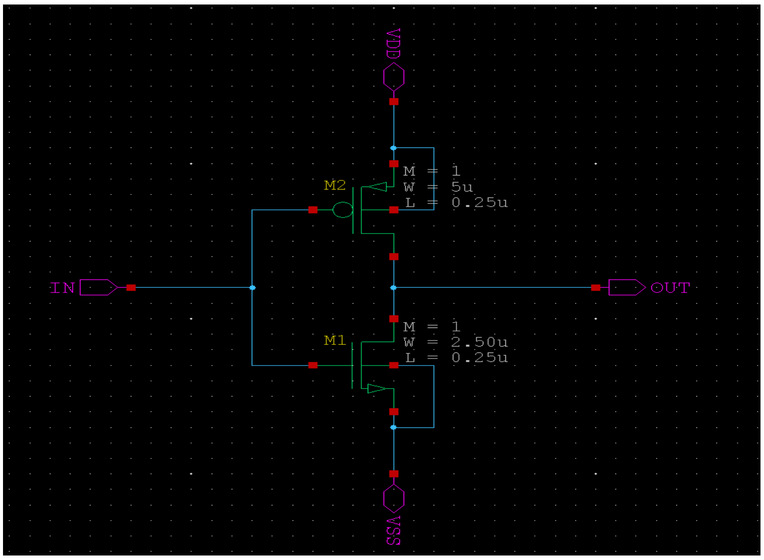

Fig 7.11: Schematic of Inverter

7.10.3Export a SPICE Netlist:

Exporting a SPICE Netlist is a good idea in order to verify that you have entered the schematic correctly. Also, this Netlist will be used later when performing a "Layout versus Schematic (LVS)" check. We want to export a Netlist at the Inverter schematiccell level so that a Netlist of just the inverter exists for LVS. When we conduct the simulation of this inverter, we will create a TOP level schematic that will have a Netlist containing ideal voltage sources. This Netlist can't be used for LVS since it contains components thatwon't be fabricated.

With the schematic open, use the pull down menus to perform:File – Export – Export SPICE.

Browse to your design directory and give the file name "Inverter.spc".Click "OK"

If you open the Inverter.spc with a text editor, you will see the following:

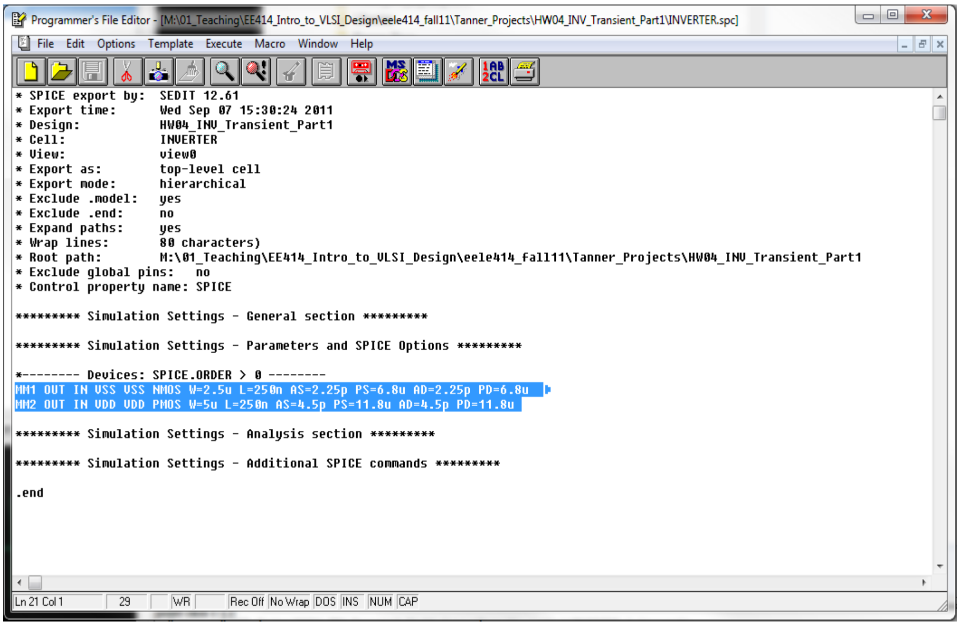

Fig 7.12: Export SPICE Netlist

7.10.4Create the Inverter Symbol:

Symbols can either be created manually by creating a new symbol view or automatically by S-edit. We will use the automatic symbol generation. This will create a new symbol view from the schematic, create the ports for the symbol, and make a symbol shape. Whilethe shape of the symbol is rarely what we ultimately want, it will do a lot of the work for us.

- With the schematic view open, use the pull down menus to create the symbol view:

Cell – Update Symbol

A new window will come up with a square symbol and 4 ports with the same names you entered in the Inverter:schematic view (i.e., IN, OUT, VDD, VSS). You should edit the shapes until you have created a symbol that looks like an inverter:

A note on drawing:

The “Path” icon will put you into a mode where you can draw lines that are not wires.The “Circle” icon will allow you to enter the inversion bubble.

The ports can be moved by holding down “alt-m”

The ports can be rotated by selecting and pressing the “r” buttonRemember to save.

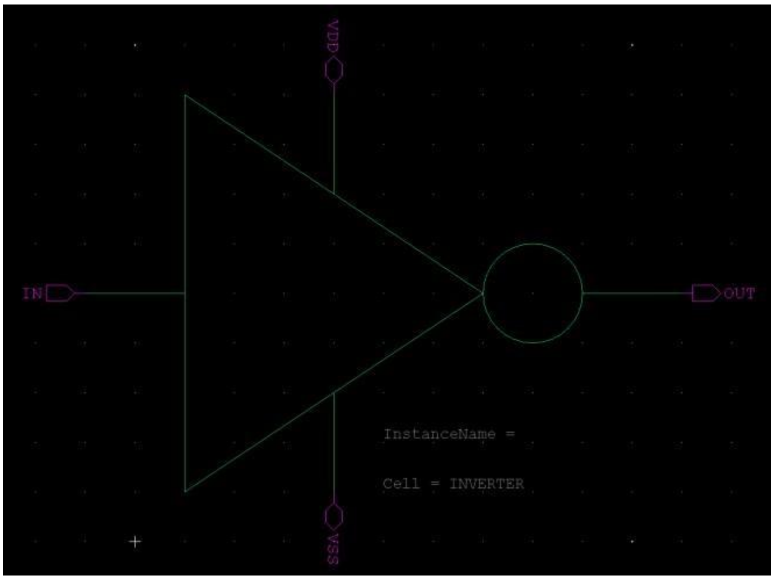

Fig 7.13: Inverter Symbol

7.11 Create the Top Schematic to Test the Inverter:

7.11.1 Instantiate the Inverter in the TOP Schematic:

Open the TOP schematic view using the pull-down menus:

- Cell – Open View:

Cell Name:	TOP View
Type:	schematic

In the library windows on the left of the window, highlight your "HW04_INV_Transient_Part1" library. In the lower left window, you will see your two Cells "TOP" and "Inverter".

- Click on "Inverter" and you will see your symbol show up in the symbol viewer.
- Click on the "Instance" button and place your symbol in the TOP schematic.

7.11.2 Power and Stimulate the Inverter:

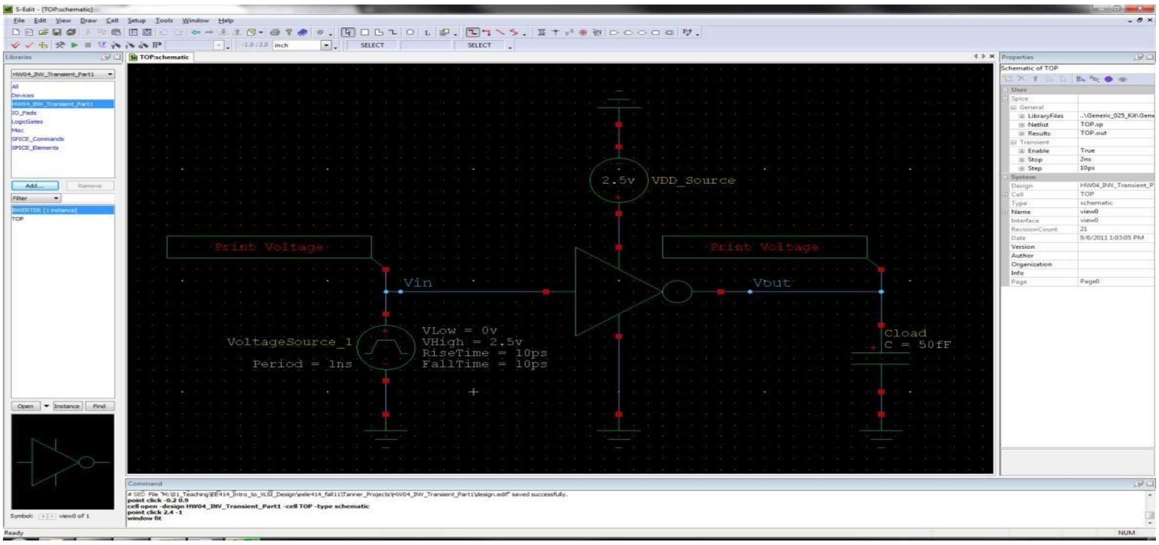

Fig 7.14: Power and Stimulate the Inverter

- Enter the Pulse Voltage Source. All voltage sources are the same component in the SPICE_Elements library. The default is DC, but this can be changed to any other type ofsource in the properties dialog.

Name	= Vin_Source
MasterInterface	= Pulse
Period	= 1ns
PulseWidth	= 0.5ns
VLow	= 0v
Risetime	= 10ps
Falltime	= 10ps

- Enter a Load capacitor from the Devices library.

Name	= Cload
C	=50fF

- Enter a DC Source for VDD

Name	= VDD_Source
MasterInterface	= DC
V	= 2.5v

- Enter the grounds from the Misc library
- Enter wire connections and name them Vin and Vout
- Enter a voltage probe for both Vin and Vout

7.12 Simulate the Design:

7.12.1 First, check you Design Using the Pull Down Menus:

- Tools – Design Checks (any warnings or errors will be shown at the bottom)

7.12.2 Simulate your design:

- Clock on the Green Arrow to start the simulator:

The T-Spice window will appear. If everything is OK, the waveform viewer will also appear. If everything worked, your waveforms should look like this:

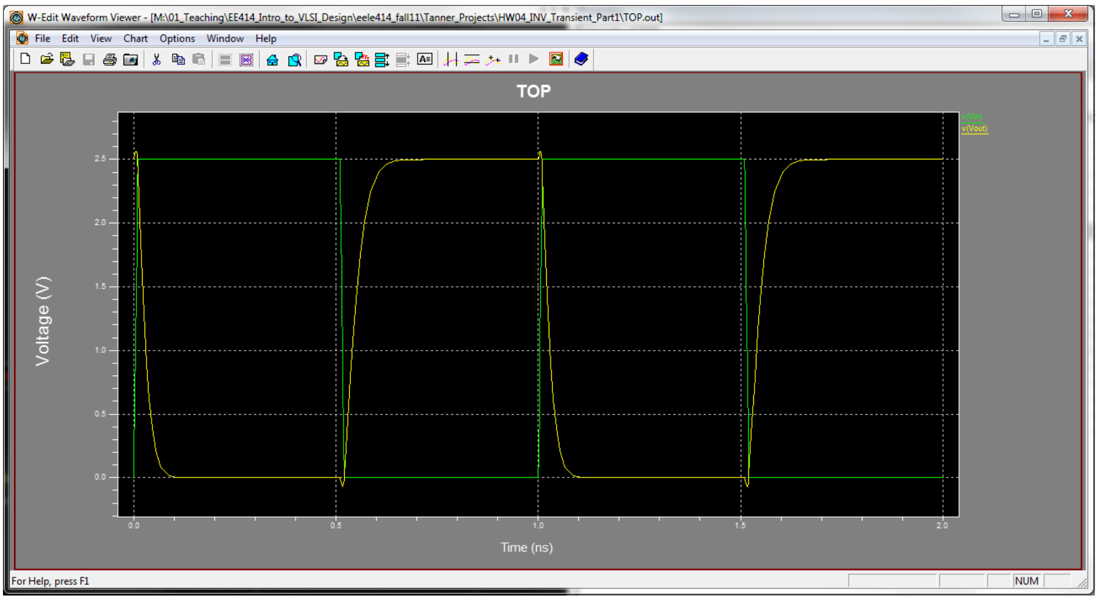

Fig 7.15: Stimulate the Design

Using L-Edit (IC Physical Design Tool)

L-edit is an integrated circuit physical design tool from Tanner EDA. This tool allows you to draw the layout of an IC, look at cross-sections, perform DRC (design rule check), and generate a Netlist of your layout so that you can perform LVS (layout versus schematic) using a different tool.

7.13 Launch L-edit, Start a Design, and Setup the Technology:

7.13.1 Log onto a computer in the Digital lab and Launch L-edit using:

Start – All Programs – Tanner EDA – Tanner Tools v12.6 – L-Edit v12.6

7.13.2 Create A New Layout Design:

- File – New
- select “Layout”
- under “Copy TDB…”, browse to:\Tanner_Projects\Generic_025_Kit\Generic_025.tdb
- Click “OK”

When you copy in the Generic_025.tbd files, it loads all of the layer definitions for the 0.25um process and the design rule information. On the left, you should see a set of layersfor this technology that can be used to create devices.

7.13.3 Verify the Technology Rule Options & Setup Grid:

- Setup – Design

Now you want to setup the grid. Click on the “Grid” tab. Since we are using a 0.25um grid, let’s put our major grid and snap grid at 0.125um. Then we will put our minor gridat 0.05. Enter the following:

7.13.4 Save your Design:

Now you can click “save” and give your design a descriptive name and location. (i.e., \Tanner_Projects\HW07_INV_Layout_Part1”

7.14 Inspect the Design Rules for the Kit:

The design rules for this kit were loaded when you specified the Generic 0.25um design kit. L-edit willcheck for violations in the design rules using a process called Design Rule Check (DRC). To see the rules, use the pull down menus:

- Tools – DRC Setup
- You’ll see that the “DRC Standard Rule Set” is selected. Highlight this (if it isn’talready) and click the“Edit” Button:

Once you look at the rules, click “OK” to go back to the “Setup DRC” dialog and checkthe “Pop upmessage box”